BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Testing and Test Data

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does anticipating misuse mean in defensive design?

2. Merge sorted [1, 5] and [4, 6]. Which result is ascending?

A) [1, 5, 4, 6]
B) [4, 6, 1, 5]
C) [6, 5, 4, 1]
D) [1, 4, 5, 6]

3. "A comparison operator must always produce a text paragraph." Correct this. Explain why.

4. A school shares files and printers on one site. "A LAN must span several countries." Correct this. Why does a LAN fit here?

Today's objectives

Explain the core ideas of testing and test data using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Normal test
A typical value within permitted range.

Boundary test
A value at the smallest or largest allowed limit.

Invalid test
A value of the right type but outside allowed rules.

Erroneous test
Input of the wrong type that the program should reject safely.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Normal test	A typical value within permitted range.
Boundary test	A value at the smallest or largest allowed limit.
Invalid test	A value of the right type but outside allowed rules.
Erroneous test	Input of the wrong type that the program should reject safely.

I Do - One worked example

Create a test table for an integer score from 0 to 100 inclusive.

1. Normal 50 should be accepted.
2. Boundary 0 and 100 should both be accepted.
3. Invalid -1 and 101 should both be rejected.
4. Erroneous 'hello' should produce a safe error, not be accepted.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Plan tests for a quantity between 1 and 10.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Choose a negative test value to check that the program rejects it.

```
fileMB = 700
if fileMB >= 0 AND fileMB <= 1000 then
print("Accepted")
else
print("Rejected")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Identify two boundary tests for valid ages 13-18. [2 marks]

Q3. Distinguish invalid and erroneous data for a numeric mark. [4 marks]

Q4. State why expected outcomes belong in a test plan. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. For range 1-10, 11 is a valid boundary value. ☐ T ☐ F
2. A typical value within permitted range. ☐ T ☐ F
3. A value at the smallest or largest allowed limit. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Planning for invalid, unexpected or malicious use and handling it safely. ☐

Q2 D - [1, 4, 5, 6] ☐

Q3 A Boolean result: true or false. ☐

Q4 A LAN connects devices in a small area, such as one school site. A WAN links networks over a large area. Internet access alone does not make a LAN a WAN. ☐

Code First - Model

```
fileMB = -1
if fileMB >= 0 AND fileMB <= 1000 then
print("Accepted")
else
print("Rejected")
endif
```

Output: Rejected

We Do - Check the model

1. An erroneous text input is 'ten'; expected outcome is rejection. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 13 and 18. ☐

Q3 Invalid: numeric 101 outside 0-100; erroneous: text 'apple' where an integer is expected. ☐

Q4 They allow actual results to be compared with requirements and faults detected. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: